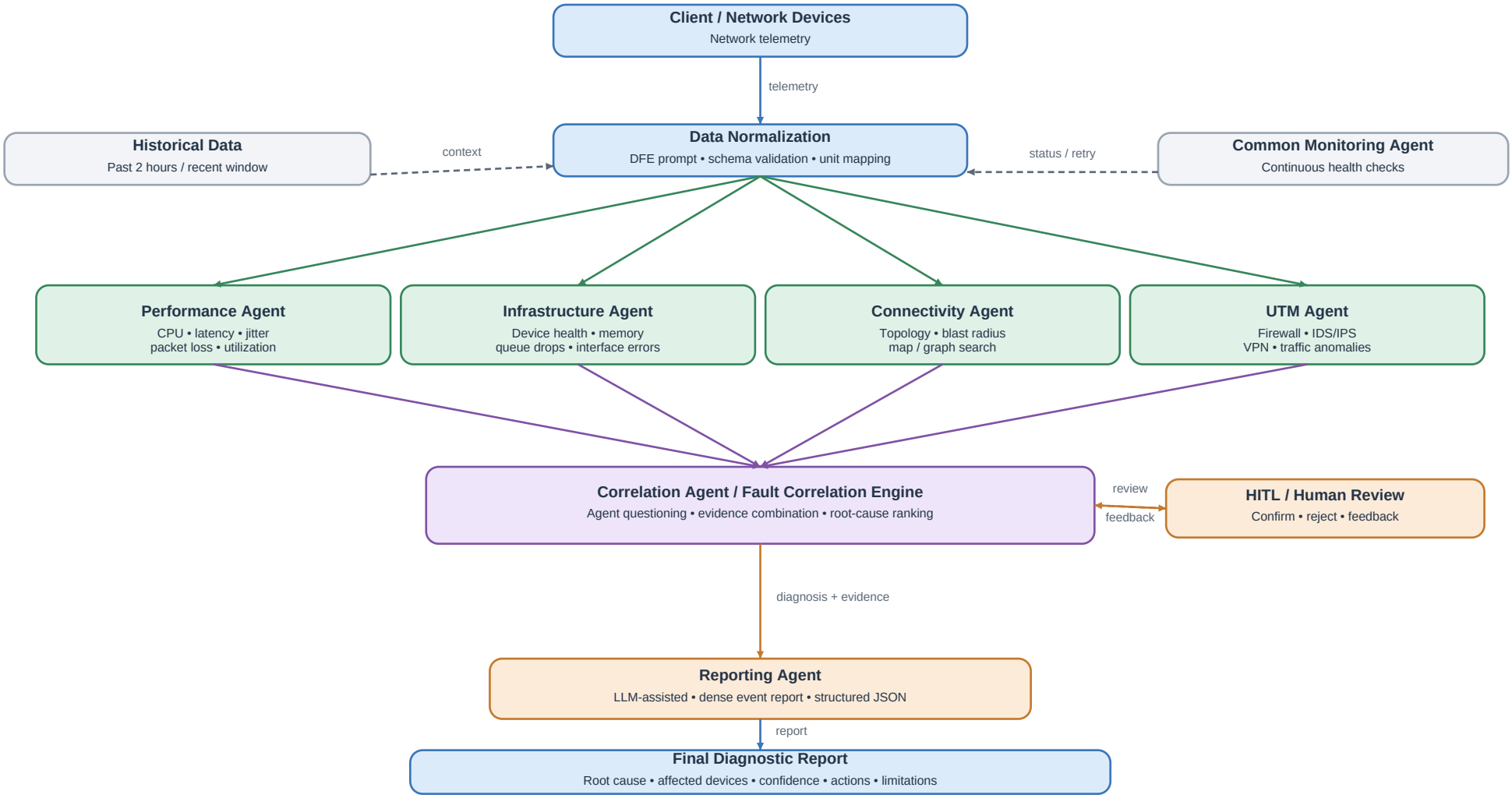
NetCoop AI

Updated multi-agent architecture with the handwritten feature ideas incorporated



How the workflow operates

1. Ingest & normalize Collect telemetry, validate it, normalize metric names and units, and attach recent historical context.	**2. Analyze** Specialist agents inspect performance, infrastructure, connectivity/topology and UTM security signals.	**3. Collaborate** Agents can question one another; the correlation engine combines evidence and handles conflicting findings.
4. Monitor & recover A common monitoring agent checks health, detects failed tasks and triggers bounded retries.	**5. Human review** HITL allows an administrator to confirm or reject the diagnosis and record feedback.	**6. Report** The reporting agent creates a detailed event report and machine-readable JSON output.

Handwritten feature ideas — incorporated

These are the additional bullet points from your updated sketch, translated into implementation-ready requirements.

- **Retries if something fails** — Add bounded retries for transient tool/API failures, with timeout limits, backoff, error logs and a clear failed/partial status. Do not retry invalid data forever.
- **Agent questioning** — Allow the correlation agent to ask a specialist agent a targeted follow-up question when evidence is missing or conflicting. Keep questions structured and limit the number of rounds.
- **LLM evaluation** — Evaluate LLM-generated explanations for evidence support, schema validity, consistency with telemetry and unsupported claims. Keep deterministic checks as the source of truth.
- **Graph with map + JSON documentation** — Show network topology as an interactive graph/map and export the topology, device relationships and incident findings as documented JSON.
- **Detailed event reporting** — Generate a time-ordered incident timeline showing metric changes, alerts, agent findings, retries, decisions and final recommendations. (Interpreted from the handwritten note 'Dense event reporting'.)
- **Common agent keeping health check** — Introduce a monitoring/supervisor agent that periodically checks agent availability, task status, stale telemetry and workflow health.

Recommended upgrades to elevate the project

- **1. Temporal anomaly detection** — Use rolling baselines and change detection to spot unusual trends, not just fixed thresholds. Example: a sudden rise in packet loss or interface errors compared with the device's own recent baseline.
- **2. Root-cause dependency graph** — Represent devices, links, services and dependencies as a graph. Combine topology with time-correlated alerts to rank likely upstream causes and show the blast radius.
- **3. Multi-fault diagnosis** — Support simultaneous faults instead of forcing every incident into one label—for example, a weak Wi-Fi signal plus an overloaded uplink.
- **4. Evidence provenance** — Every claim in the report should link back to the exact telemetry record, timestamp, device and agent that produced it. This makes the diagnosis auditable.
- **5. Confidence calibration and abstention** — Calibrate confidence against labelled scenarios. If evidence is weak, missing or contradictory, explicitly return 'insufficient evidence' rather than guessing.
- **6. Counterfactual / what-if analysis** — Show how the diagnosis would change if a suspected link/device were healthy, or simulate the impact of a proposed fix before any real action.
- **7. Safe remediation recommendations** — Offer ranked, reversible actions with expected impact and verification steps. Keep real network changes behind explicit human approval; demo with simulation only.
- **8. Benchmark and evaluation dashboard** — Show per-fault precision/recall, false alarms, agent success rate, end-to-end latency, schema compliance and performance on malformed inputs.

Implementation guardrails

- Use a shared versioned message schema for all agents; include incident ID, timestamp, source, units, status, findings and limitations.
- Keep retries bounded and idempotent. Retry transient failures, not bad input or deterministic validation errors.
- Treat the UTM agent as an optional security extension when firewall/IDS/IPS/VPN logs are available; otherwise use clearly labelled synthetic data.
- Use LLMs to explain and question evidence, not to invent telemetry or override deterministic validation.
- Build the end-to-end minimum workflow first. Prioritize reliable interoperability and diagnosis before advanced features.

Suggested hackathon priority: first deliver a working baseline with structured agent messages, three or more functional diagnostic agents, correlation, an auditable report and malformed-input handling. Add the upgrades selectively after the baseline is stable.